

Not Finished, But Abandoned

by

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ACKNOWLEDGMENTS

It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....

Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.

— DAVID C. MAKINSON (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

CONTENTS

Contents ii

List of Tables viii

List of Figures ix

Abstract x

1 Introduction 1

1.1	<i>What the heck is a plasma</i>	1
1.1.1	Plasmas in the universe	1
1.1.2	How do plasmas affect us on earth?	1
1.1.3	Where does magnetic reconnection play into this whole shabang?	1
1.2	<i>Magnetic Reconnection</i>	1
1.2.1	The Generalized Ohm's Law	1
1.2.1.1	How do we derive this?	1
1.2.1.2	A breakdown of the terms	1
1.2.2	Regimes of Reconnection	1
1.2.2.1	Reconnection scaling parameters	1
1.2.2.2	Where do various systems line up?	2
1.2.2.3	What are we interested in?	2
1.2.3	Sweet-Parker Reconnection	2
1.2.3.1	A quick historical detour	2
1.2.3.2	What are the assumptions that go into Sweet-Parker?	2
1.2.3.3	Sweet-Parker Reconnection Rate Derivation	2
1.2.3.4	Application to Real Systems	2

1.2.3.4.1	What are some example parameters where reconnection occurs? .	2
1.2.3.4.2	What time scales does Sweet-Parker predict?	2
1.2.3.4.3	What did we miss?	2
1.2.4	Hall Reconnection	2
1.2.4.1	Why should the Hall term appear?	2
1.2.4.1.1	What is the relevant physics that sets when the hall term matters .	2
1.2.4.1.2	In what kind of plasmas does the hall term appear and when does it not?	2
1.2.5	Kinetic Reconnection	3
1.2.5.1	What makes kinetic reconnection “kinetic”? .	3
1.2.5.1.1	What do collisions do?	3
1.2.5.1.2	What makes a system kinetic? . .	3
1.2.5.2	Where does Pressure Anisotropy come from? .	3
1.2.5.2.1	Movement of flux tube	3
1.2.5.2.2	Effect of background field	3
1.2.5.3	How does pressure anisotropy change what the reconnection looks like	4
1.2.5.3.1	Thinner reconnection layer?	4
1.2.5.3.2	Marginal Firehose Condition . . .	4
1.2.5.3.3	Concluding: what features should we look for in the data?	4
1.2.5.4	Why do we want to measure it?	4
1.3	<i>Experimental Setup</i> 6	
1.3.1	Overall how does our experiment progress	6
1.3.2	Initial field	6
1.3.2.1	Helmholtz	6

1.3.2.2	Permanent magnets	6
1.3.2.3	Toroidal	6
1.3.2.3.1	Shifted coil	6
1.3.2.3.2	Axial coil	6
1.3.2.4	Perturbation coils	6
1.3.3	Plasma Generation	6
1.3.3.1	How do our guns work	6
1.3.3.1.1	Gas lines	6
1.3.3.1.2	PFNs	6
1.3.3.1.3	Puff trigger	6
1.3.3.1.4	How to breakdown	6
1.3.3.2	How are the guns placed for injection? . .	7
1.3.3.2.1	How much do they fill the machine	7
1.3.3.2.2	What sits the filling fraction	7
1.3.3.3	What goes wrong with them	7
1.3.3.3.1	Burning up	7
1.3.3.3.2	Burn holes in wall because of back- ground field	7
1.3.3.3.3	Air cooling	7
1.3.3.3.4	Optimizing gun operation	7
1.3.4	Drive field	7
1.3.4.1	Individual coils	7
1.3.4.1.1	Zone's between coils as plasma sources	7
1.3.4.2	Drive Cylinder	7
1.3.5	Diagnostics Overview	8
1.3.5.1	Bdot probes	8
1.3.5.1.1	Basic principles	8
1.3.5.1.2	Probe types	8
1.3.5.1.3	Data Interpretation	8
1.3.5.2	Hall probes	8

1.3.5.2.1	Basic principles	8
1.3.5.2.2	Issues and feasibility	8
1.3.5.3	Langmuir probes	8
1.3.5.3.1	Basic principles	8
1.3.5.3.2	What do we need to do to make them work for our device?	8
1.3.5.3.3	Look into XXX chapter	8
1.3.5.4	Fast camera	9
1.3.5.4.1	What can we see with it?	9
1.3.5.4.2	Potential use cases	9
1.3.6	How does the experiment compare to space	9
1.3.6.1	Benefits of lab vs. space vs. simulation . .	9
1.3.6.1.1	Simulation benefits and results . .	9
1.3.6.1.2	Space benefits and results	9
1.3.6.1.3	Lab benefits and results	9
1.3.7	Other lab reconnection experiments	9
1.3.7.1	VTF, MRX, FLARE	9
1.3.7.2	Exploding wire arrays	9
1.3.7.3	Laser based	9
2	Pressure Wave 10	
2.1	<i>Introduction</i> 10	
2.2	<i>Observations that lead us to look at this wave</i> 11	
2.2.1	Speed probe measurements	11
2.2.1.1	Flux expansion before shock layer	11
2.2.1.2	Wavefront that depends on configuration .	11
2.2.2	Simulations	11
2.2.2.1	Different n and drive change feature	11
2.2.2.2	Change reconnection layer post reflection .	11
2.3	<i>Physical understanding of wave</i> 11	
2.3.1	Initial plasma configuration	11

2.3.2	Release in magnetic pressure	11
2.3.2.1	What would happen if plasma was im- mobile (replaced with perfect conductor) . .	11
2.3.3	Wavefront tells plasma about pressure Release . . .	11
2.3.3.1	Analogous system	11
2.4	<i>Wave Simulation</i> 12	
2.4.1	Approximations	12
2.4.2	Linearizing MHD	12
2.4.3	Plasma Equilibrium + Symmetries	12
2.4.3.1	Types of Equilibrium	12
2.4.3.1.1	theta-pinch	12
2.4.3.1.2	screw-pinch	12
2.4.3.2	Types of Symmetries	12
2.4.3.2.1	z-symmetry	12
2.4.3.2.2	toroidal-symmetry	12
2.4.4	Simplified 1D equation	12
2.4.4.1	What does this look like?	12
2.4.4.2	Normalizations	13
2.4.4.3	Normalized equation	13
2.4.5	Boundary conditions	13
2.4.5.1	$r = 0$ Boundary	13
2.4.5.1.1	$v = 0$ option	13
2.4.5.1.2	$\xi = 0$ option	13
2.4.5.1.3	What do we choose?	13
2.4.5.2	$r = R$ Boundary	13
2.4.5.2.1	What is physically happening at high radius in the coils?	13
2.4.5.2.2	Loop voltage sets dB_z/dt	13
2.4.5.2.3	Assuming MHD so far so let's keep assuming!	13

2.4.5.2.4	Vloop signal shape	13
2.4.6	Initial conditions	13
2.4.6.1	What form can we expect plasma to take? .	13
2.4.6.2	What form do we use for these results . . .	13
2.4.7	What does our simulation show?	13
2.4.7.1	There is a wavefront traveling inwards at fast magnetosonic	13
2.4.7.1.1	Low beta approximation giving Alfven	13
2.4.7.2	Once wavefront passes, plasma moves out	14
2.4.7.3	Converting to B_{dot}	14
2.4.7.4	In B_{dot} we see flux moving outwards . . .	14
3	Multi-tip Langmuir Probe	15
4	Pressure Anisotropy Probe	16
5	Conclusion	17
	Colophon	18

LIST OF TABLES

1.1	List of diagnostics used in the experiment and what they can measure.	7
1.2	Lab vs. Space Parameters	9
2.1	Parameters for normalization	12

LIST OF FIGURES

1.1	Cartoon of places where reconnection can happen.	1
1.2	Diagram of reconnection regimes.	1
1.3	Diagram of Sweet-Parker reconnection.	2
1.4	Diagram of 2d hall reconnection	2
1.5	Flux tube changing size during reconnection causing pressure anisotropy build up.	3
1.6	Simulated kinetic reconnection with changing guide field. . .	3
1.7	Diagram from Ohia of with and without pressure anisotropy in simulation.	4
1.8	Some figure from Jan's papers thowing effects of pressure anisotropy	4
1.9	Timeline of experimental sequence.	5
1.10	Panels from flux arrays of experiment happening.	5
1.11	Circuit diagram of plasma gun.	6
1.12	Simplified cut of plasma gun insides.	6
1.13	Image from high speed camera of gun spewing	7
2.1	dB_z/dt plots for different Helmholtz fields	10
2.2	Greess et al. 2022 paper showing simulations with pressure wave.	10

ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,
2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

1 INTRODUCTION

1.1 What the heck is a plasma

1.1.1 Plasmas in the universe

1.1.2 How do plasmas affect us on earth?

1.1.3 Where does magnetic reconnection play into this whole shabang?

1.2 Magnetic Reconnection

1.2.1 The Generalized Ohm's Law

1.2.1.1 How do we derive this?

1.2.1.2 A breakdown of the terms

1.2.2 Regimes of Reconnection

1.2.2.1 Reconnection scaling parameters

i.e. why do we use S and L/d_i to characterize reconnection regimes?

Figure 1.1: Cartoon of places where reconnection can happen.

Figure 1.2: Diagram of reconnection regimes.

Figure 1.3: Diagram of Sweet-Parker reconnection.

Figure 1.4: Diagram of 2d hall reconnection

1.2.2.2 Where do various systems line up?

1.2.2.3 What are we interested in?

1.2.3 Sweet-Parker Reconnection

1.2.3.1 A quick historical detour

1.2.3.2 What are the assumptions that go into Sweet-Parker?

1.2.3.3 Sweet-Parker Reconnection Rate Derivation

1.2.3.4 Application to Real Systems

1.2.3.4.1 What are some example parameters where reconnection occurs?

1.2.3.4.2 What time scales does Sweet-Parker predict?

1.2.3.4.3 What did we miss?

1.2.4 Hall Reconnection

1.2.4.1 Why should the Hall term appear?

1.2.4.1.1 What is the relevant physics that sets when the hall term matters

1.2.4.1.2 In what kind of plasmas does the hall term appear and when does it not?

Figure 1.5: Flux tube changing size during reconnection causing pressure anisotropy build up.

Figure 1.6: Simulated kinetic reconnection with changing guide field.

1.2.5 Kinetic Reconnection

1.2.5.1 What makes kinetic reconnection “kinetic”?

1.2.5.1.1 What do collisions do?

1.2.5.1.1.1 How quickly do collisions damp out non-Maxwellian features?

1.2.5.1.2 What makes a system kinetic?

1.2.5.1.2.1 What parameters set this?

1.2.5.1.2.2 What should we do in the lab to get this?

1.2.5.2 Where does Pressure Anisotropy come from?

1.2.5.2.1 Movement of flux tube

1.2.5.2.2 Effect of background field

1.2.5.2.2.1 Too low of background field does...

1.2.5.2.2.2 Too high of field does...

1.2.5.2.2.3 Goldilocks!

Figure 1.7: Diagram from Ohia of with and without pressure anisotropy in simulation.

Figure 1.8: Some figure from Jan's papers showing effects of pressure anisotropy

1.2.5.3 How does pressure anisotropy change what the reconnection looks like

1.2.5.3.1 Thinner reconnection layer? Use results from Jan's more recent papers.

1.2.5.3.2 Marginal Firehose Condition

1.2.5.3.3 Concluding: what features should we look for in the data?

1.2.5.4 Why do we want to measure it?

At the end of experimental setup I'll talk about why we want to do it in the lab but I think motivating why we care about kinetic reconnection as a process would be good here.

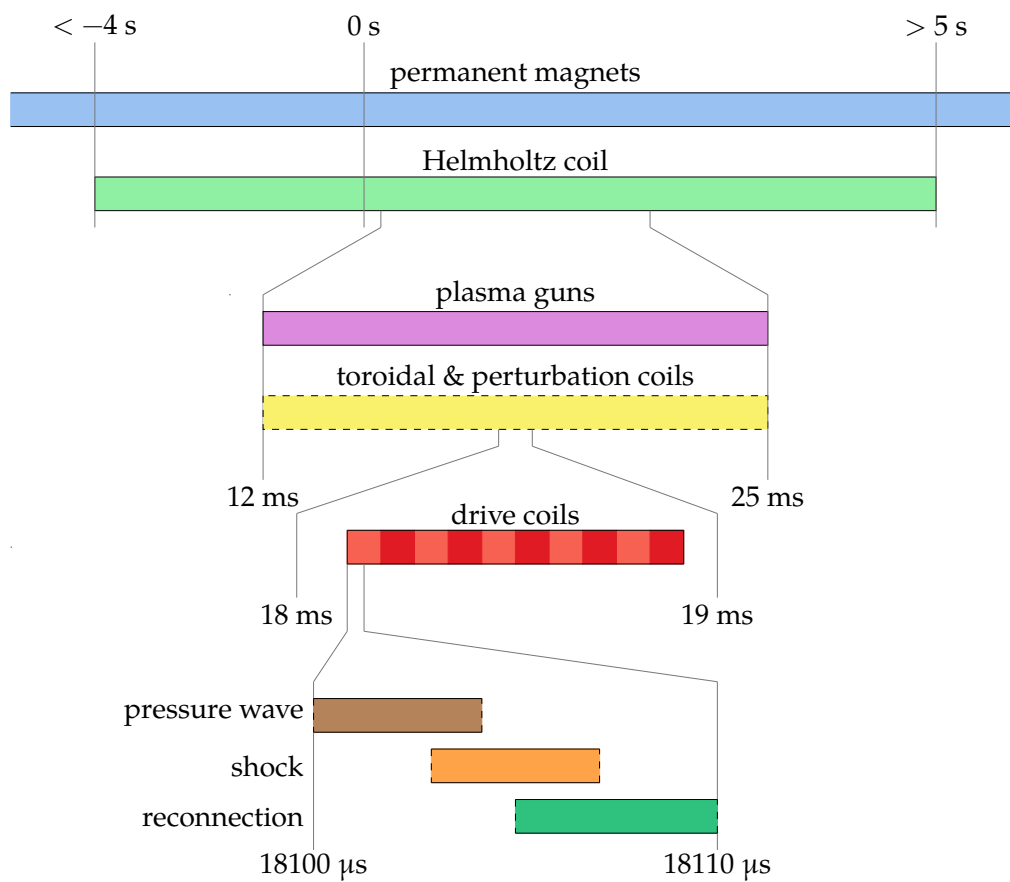


Figure 1.9: Timeline of experimental sequence.

Figure 1.10: Panels from flux arrays of experiment happening.

Figure 1.11: Circuit diagram of plasma gun.

Figure 1.12: Simplified cut of plasma gun insides.

1.3 Experimental Setup

1.3.1 Overall how does our experiment progress

1.3.2 Initial field

1.3.2.1 Helmholtz

1.3.2.2 Permanent magnets

1.3.2.3 Toroidal

1.3.2.3.1 Shifted coil

1.3.2.3.2 Axial coil

1.3.2.4 Perturbation coils

1.3.3 Plasma Generation

1.3.3.1 How do our guns work

1.3.3.1.1 Gas lines

1.3.3.1.2 PFNs

1.3.3.1.3 Puff trigger

1.3.3.1.4 How to breakdown

Figure 1.13: Image from high speed camera of gun spewing

Table 1.1: List of diagnostics used in the experiment and what they can measure.

1.3.3.2 How are the guns placed for injection?

1.3.3.2.1 How much do they fill the machine

1.3.3.2.2 What sits the filling fraction

1.3.3.3 What goes wrong with them

1.3.3.3.1 Burning up

1.3.3.3.2 Burn holes in wall because of background field Get image from Paul with hole in wall.

1.3.3.3.3 Air cooling

1.3.3.3.4 Optimizing gun operation

1.3.4 Drive field

1.3.4.1 Individual coils

1.3.4.1.1 Zone's between coils as plasma sources

1.3.4.2 Drive Cylinder

Reference Paul's RSI + thesis.

1.3.5 Diagnostics Overview

1.3.5.1 Bdot probes

1.3.5.1.1 Basic principles

1.3.5.1.2 Probe types

1.3.5.1.2.1 Linear probe

1.3.5.1.2.2 Hook probe

1.3.5.1.2.3 Speed probe

1.3.5.1.2.4 Flux array 1

1.3.5.1.2.5 Flux array 2

1.3.5.1.3 Data Interpretation How do we convert the raw data into something useful?

1.3.5.2 Hall probes

1.3.5.2.1 Basic principles

1.3.5.2.2 Issues and feasibility

1.3.5.3 Langmuir probes

1.3.5.3.1 Basic principles

1.3.5.3.2 What do we need to do to make them work for our device?

1.3.5.3.3 Look into XXX chapter

Table 1.2: Comparison of lab and space parameters.

1.3.5.4 Fast camera

1.3.5.4.1 What can we see with it?

1.3.5.4.2 Potential use cases

1.3.6 How does the experiment compare to space

1.3.6.1 Benefits of lab vs. space vs. simulation

1.3.6.1.1 Simulation benefits and results

1.3.6.1.2 Space benefits and results

1.3.6.1.3 Lab benefits and results

1.3.7 Other lab reconnection experiments

1.3.7.1 VTF, MRX, FLARE

1.3.7.2 Exploding wire arrays

1.3.7.3 Laser based

2 PRESSURE WAVE

2.1 Introduction

Once I get Jan's rewrite of the intro + conclusion I can figure out what might go here.

Figure 2.1: dBz/dt plots for different Helmholtz fields

Figure 2.2: Greess et al. 2022 paper showing simulations with pressure wave.

2.2 Observations that lead us to look at this wave

2.2.1 Speed probe measurements

2.2.1.1 Flux expansion before shock layer

2.2.1.2 Wavefront that depends on configuration

2.2.2 Simulations

2.2.2.1 Different n and drive change feature

2.2.2.2 Change reconnection layer post reflection

2.3 Physical understanding of wave

2.3.1 Initial plasma configuration

2.3.2 Release in magnetic pressure

2.3.2.1 What would happen if plasma was immobile (replaced with perfect conductor)

All field is shielded at edge and there is no change in magnetic pressure in the bulk.

2.3.3 Wavefront tells plasma about pressure Release

2.3.3.1 Analogous system

Sound wave.

How does a sound wave propagate in a plasma? Fast magnetosonic wave!

Table 2.1: Parameters for normalization

2.4 Wave Simulation

2.4.1 Approximations

2.4.2 Linearizing MHD

2.4.3 Plasma Equilibrium + Symmetries

2.4.3.1 Types of Equilibrium

2.4.3.1.1 theta-pinch

2.4.3.1.2 screw-pinch

2.4.3.2 Types of Symmetries

2.4.3.2.1 z-symmetry

2.4.3.2.2 toroidal-symmetry

2.4.4 Simplified 1D equation

2.4.4.1 What does this look like?

A wave equation!

2.4.4.2 Normalizations

2.4.4.3 Normalized equation

2.4.5 Boundary conditions

2.4.5.1 $r = 0$ Boundary

2.4.5.1.1 $v = 0$ option

2.4.5.1.2 $\xi = 0$ option

2.4.5.1.3 What do we choose?

2.4.5.2 $r = R$ Boundary

2.4.5.2.1 What is physically happening at high radius in the coils?

2.4.5.2.2 Loop voltage sets dB_z/dt

2.4.5.2.3 Assuming MHD so far so let's keep assuming!

2.4.5.2.4 Vloop signal shape

2.4.6 Initial conditions

2.4.6.1 What form can we expect plasma to take?

2.4.6.2 What form do we use for these results

2.4.7 What does our simulation show?

2.4.7.1 There is a wavefront traveling inwards at fast magnetosonic

2.4.7.1.1 Low beta approximation giving Alfven

2.4.7.2 Once wavefront passes, plasma moves out

2.4.7.3 Converting to B_{dot}

2.4.7.4 In B_{dot} we see flux moving outwards

As expected from MHD assumption.

3 MULTI-TIP LANGMUIR PROBE

4 PRESSURE ANISOTROPY PROBE

5 CONCLUSION

DISCARD THIS PAGE

COLOPHON

This template uses Gyre Pagella by default. (I used Arno Pro in my dissertation.)

Feel free to give me a shout-out in your colophon or acks if this template is useful for you. Good luck!